

A Closed Loop Intent Based Agent for Resilient Multi-UPF 5G Networks

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Abstract—We present a Closed Loop Intent Based Agent for autonomous resilience in 5G Core networks. By leveraging the Uplink Classifier (ULCL) and a custom Dynamic Steering API, our system detects “gray failure” on the N9 interface and triggers real-time path relocation. As gray failure we define a subtle performance degradation, such as high packet loss, occurring while a network link remains technically active and seemingly operational. Experimental results using free5GC [1] demonstrate that our agent can restore a failing video service from a Mean Opinion Score (MOS) of 2.0 to 3.5 in under 10 seconds, proving the viability of self-healing softwarized user planes.

I. SYSTEM DESIGN AND METHODOLOGY

Fig. 1 illustrates the system architecture, where a single Branching UPF connects via independent N9 tunnels to two separate Anchor UPFs [2]. By leveraging this multi-path topology, a closed loop intent based agent monitors link health and dynamically steers data flows without session interruption. By steering traffic between UPF anchors, the system achieves seamless self-healing without interrupting service continuity.

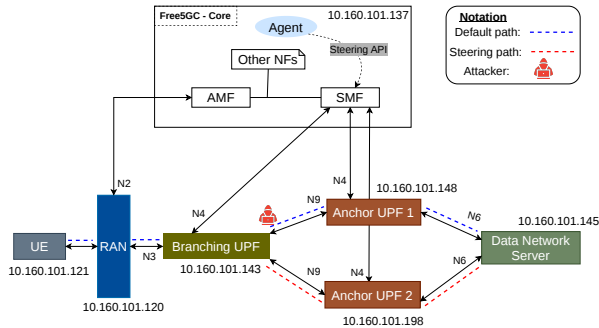


Fig. 1. System Architecture with Multi-UPF Branching.

II. EXPERIMENTAL RESULTS AND ANALYSIS

Testing utilized a high-definition RTP video stream (FFmpeg) from the UE to the Data Network server. A three-phase methodology was employed: (1) establishing a baseline via the primary Anchor UPF, (2) injecting a gray failure at the Branching UPF using `tc-netem` (22%–97% packet loss and 15ms latency), and (3) autonomous healing via a closed loop agent. The agent monitors N9 integrity through periodic probing, steering traffic to the secondary Anchor UPF once a predefined threshold is surpassed. This threshold corresponds to 10% packet loss, which was selected to provide an “acceptable”



Fig. 2. Visual impact of an N9 gray failure.

service quality level of 3.50 MOS [3]. Fig. 2 compares the original video stream against the degraded output. Free5GC does not include built-in traffic steering capabilities [1].

As detailed in Table I, the autonomous agent successfully restored service continuity across varying failure intensities (represented by 22% and 97% packet loss levels). During the Phase 2 gray failure, packet loss escalated to between 18.7% and 47.43%, causing service degradation to a “Fair” or “Poor” rating (MOS 2.00–3.00). Following autonomous steering in Phase 3, the agent mitigated loss to a range of 6.47%–8.97% and stabilized network performance, consistently restoring the service to a “Good” rating (3.50 MOS). The total control loop latency was 6.94 s for the 22% scenario and 9.23 s for the 97% packet loss scenario. Failure detection and confirmation were 4.51 s and 6.04 s for the first and second scenarios, with corresponding PFCP rule application of 1.42 s and 3.20 s.

TABLE I
SYSTEM PERFORMANCE COMPARISON

Metric	Phase 1: Baseline		Phase 2: Gray Failure		Phase 3: Recovery	
	22%	97%	22%	97%	22%	97%
Packet Loss (%)	0.04%	0.13%	18.7%	47.43%	8.97%	6.47%
Avg. Inter-arrival	7.45 ms	4.53 ms	2.097 ms	1.13 ms	1.74 ms	1.56 ms
Estimated MOS	4.50 (Exc.)	4.50 (Exc.)	3.00 (Fair)	2.00 (Poor)	3.50 (Good)	3.50 (Good)

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